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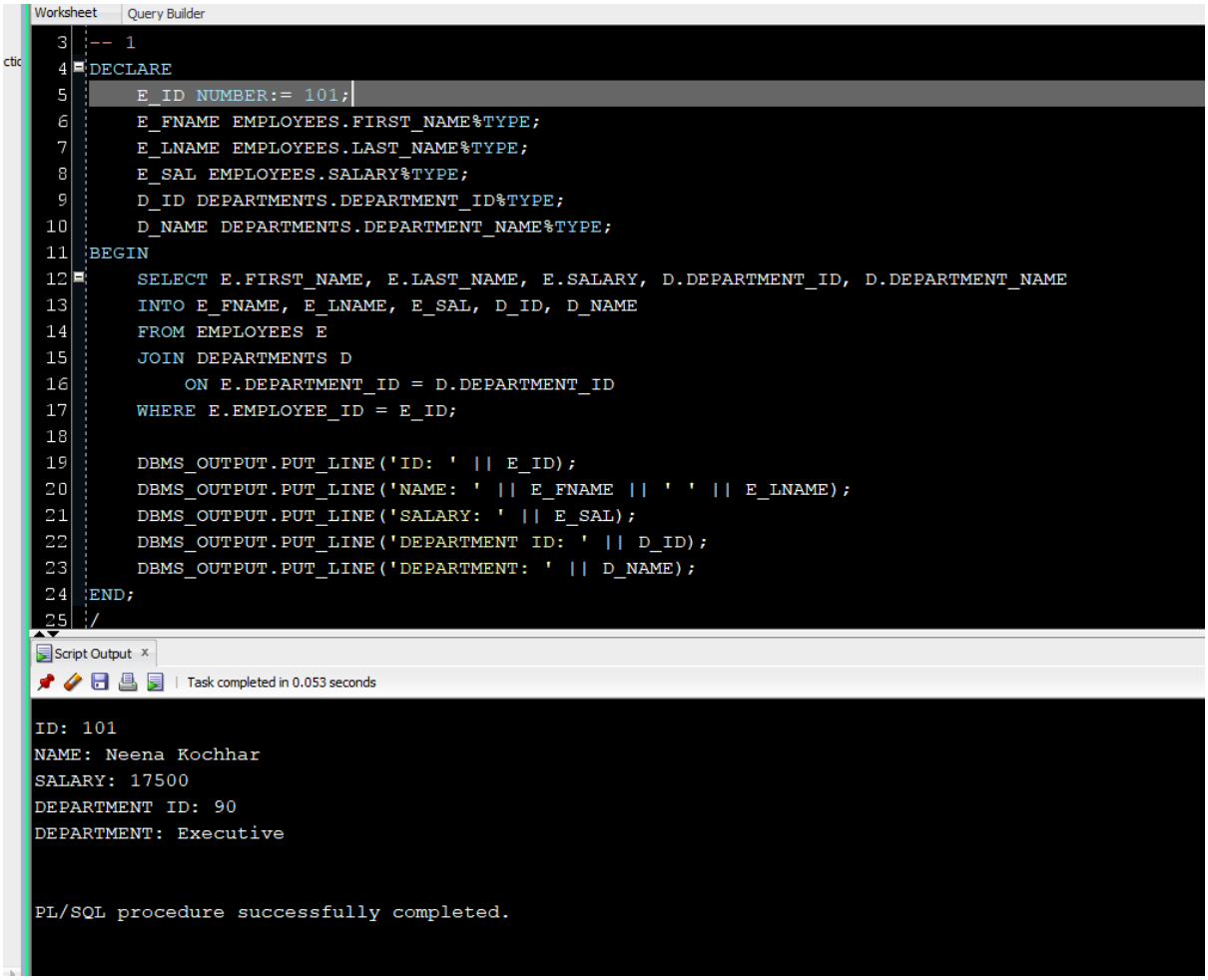
DBS Lab 10 Tasks

1)



2)

```
-- 2
BEGIN
    FOR C IN (
        SELECT FIRST_NAME, LAST_NAME, JOB_ID, SALARY
        FROM EMPLOYEES)
    LOOP
        DBMS_OUTPUT.PUT(C.FIRST_NAME || ' ' || C.LAST_NAME || ': ');
        IF C.JOB_ID LIKE '%PROG%' OR C.JOB_ID LIKE '%ENG%' THEN
            IF C.SALARY > 5000 THEN
                DBMS_OUTPUT.PUT_LINE('Eligible for Technical Recognition');
            ELSE
                DBMS_OUTPUT.PUT_LINE('Salary below threshold.');
            END IF;
        ELSE
            DBMS_OUTPUT.PUT_LINE('Standard Classification');
        END IF;
    END LOOP;
END;
/
```

Script Output

Task completed in 0.062 seconds

```
Donald OConnell: Standard Classification
Douglas Grant: Standard Classification
Jennifer Whalen: Standard Classification
Michael Hartstein: Standard Classification
Pat Fay: Standard Classification
Susan Mavris: Standard Classification
Hermann Baer: Standard Classification
Shelley Higgins: Standard Classification
William Gietz: Standard Classification


PL/SQL procedure successfully completed.
```

3)

```
47  -- 3
48  BEGIN
49      FOR C IN (
50          SELECT FIRST_NAME, LAST_NAME, SALARY, JOB_ID
51          FROM EMPLOYEES)
52          LOOP
53              DECLARE
54                  SAL_LVL VARCHAR2(10) :=
55                      CASE
56                          WHEN C.SALARY > 10000 THEN 'High'
57                          ELSE 'Standard'
58                      END;
59                  PERFORM VARCHAR2(50) :=
60                      CASE
61                          WHEN C.JOB_ID LIKE '%MAN%' THEN 'Strategic'
62                          ELSE 'Operational'
63                      END;
64              BEGIN
65                  DBMS_OUTPUT.PUT_LINE(C.FIRST_NAME || ' ' || C.LAST_NAME ||
66                  ' | Level: ' || SAL_LVL || ' | Label: ' || PERFORM);
67              END;
68          END LOOP;
69  END;
```

Script Output

Task completed in 0.048 seconds

```
Susan Mavris | Level: Standard | Label: Operational
Hermann Baer | Level: Standard | Label: Operational
Shelley Higgins | Level: High | Label: Operational
William Gietz | Level: Standard | Label: Operational


PL/SQL procedure successfully completed.
```

4)

```
-- 4
BEGIN
    FOR C IN (
        SELECT LAST_NAME, SALARY, NVL(COMMISSION_PCT, 0) AS COMM
        FROM EMPLOYEES)
        LOOP
            DBMS_OUTPUT.PUT(C.LAST_NAME || ' Status: ');
            IF (C.SALARY * C.COMM) > 1000 OR C.SALARY > 12000 THEN
                DBMS_OUTPUT.PUT_LINE('High Perfomer');
            ELSE
                DBMS_OUTPUT.PUT_LINE('Standard Performer');
            END IF;
        END LOOP;
END;
/
```

Script Output

Task completed in 0.053 seconds

```
Grant Status: Standard Performer
Whalen Status: Standard Performer
Hartstein Status: High Perfomer
Fay Status: Standard Performer
Mavris Status: Standard Performer
Baer Status: Standard Performer
Higgins Status: High Perfomer
Gietz Status: Standard Performer


PL/SQL procedure successfully completed.
```

5)

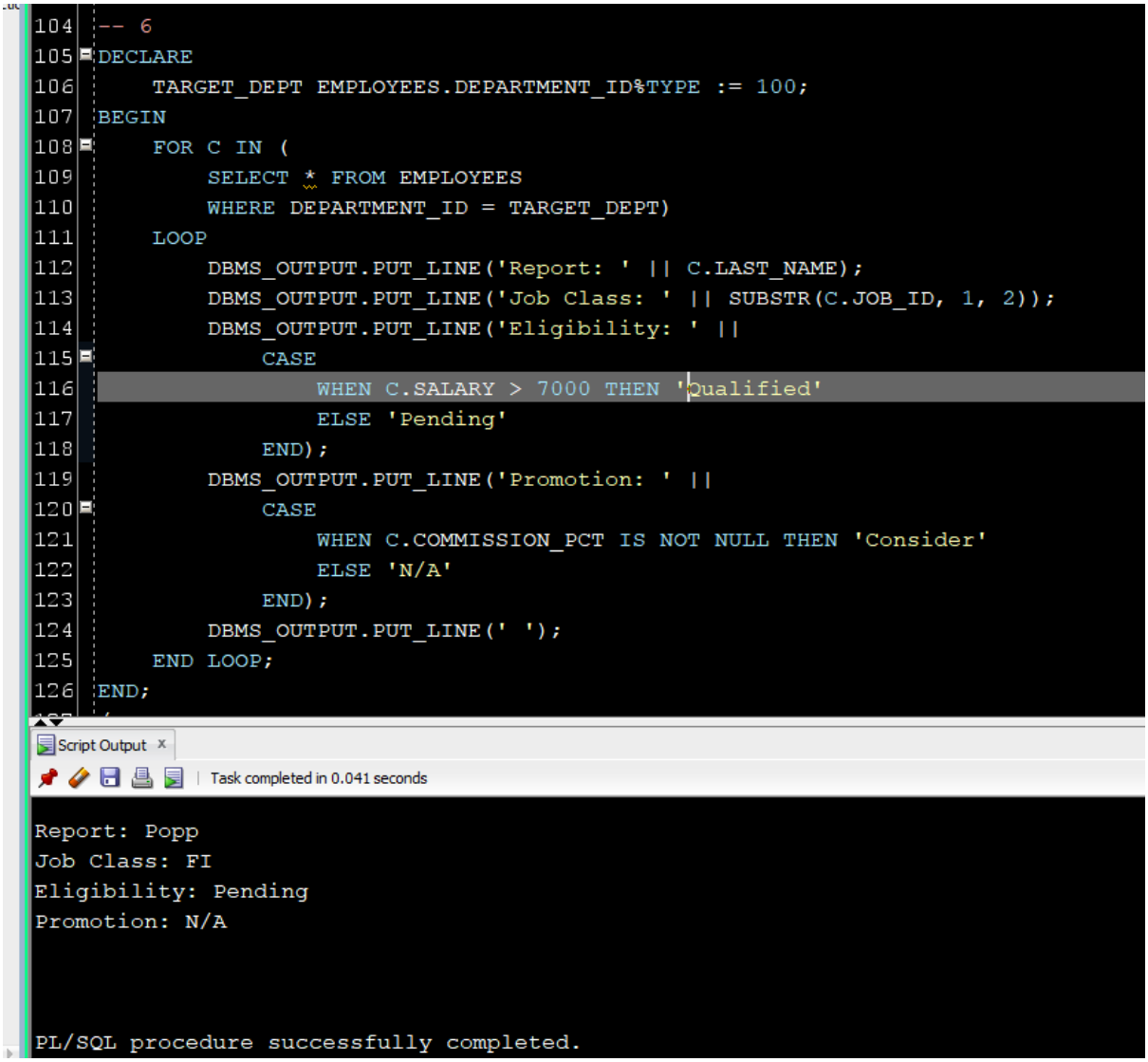
```
-- 5
BEGIN
    FOR C IN (
        SELECT NVL(DEPARTMENT_ID, 0) AS DEPARTMENT_ID, SUM(SALARY) AS TOTAL_SAL, AVG(SALARY) AS AVG_SAL
        FROM EMPLOYEES GROUP BY DEPARTMENT_ID)
    LOOP
        DBMS_OUTPUT.PUT('Dept ' || C.DEPARTMENT_ID || ': ');
        IF C.TOTAL_SAL > 50000 THEN
            DBMS_OUTPUT.PUT_LINE('Over Budget (Total: ' || C.TOTAL_SAL || ')');
        ELSE
            DBMS_OUTPUT.PUT_LINE('Normal Budget: (Avg: ' || ROUND(C.AVG_SAL,2) || ')');
        END IF;
    END LOOP;
END;
/
```

Script Output

Task completed in 0.052 seconds

```
Dept 20: Normal Budget: (Avg: 9500)
Dept 70: Normal Budget: (Avg: 10000)
Dept 110: Normal Budget: (Avg: 10154)
Dept 50: Over Budget (Total: 156400)
Dept 80: Over Budget (Total: 304500)
Dept 40: Normal Budget: (Avg: 6500)
Dept 60: Normal Budget: (Avg: 5760)
Dept 10: Normal Budget: (Avg: 4400)


PL/SQL procedure successfully completed.
```

6)



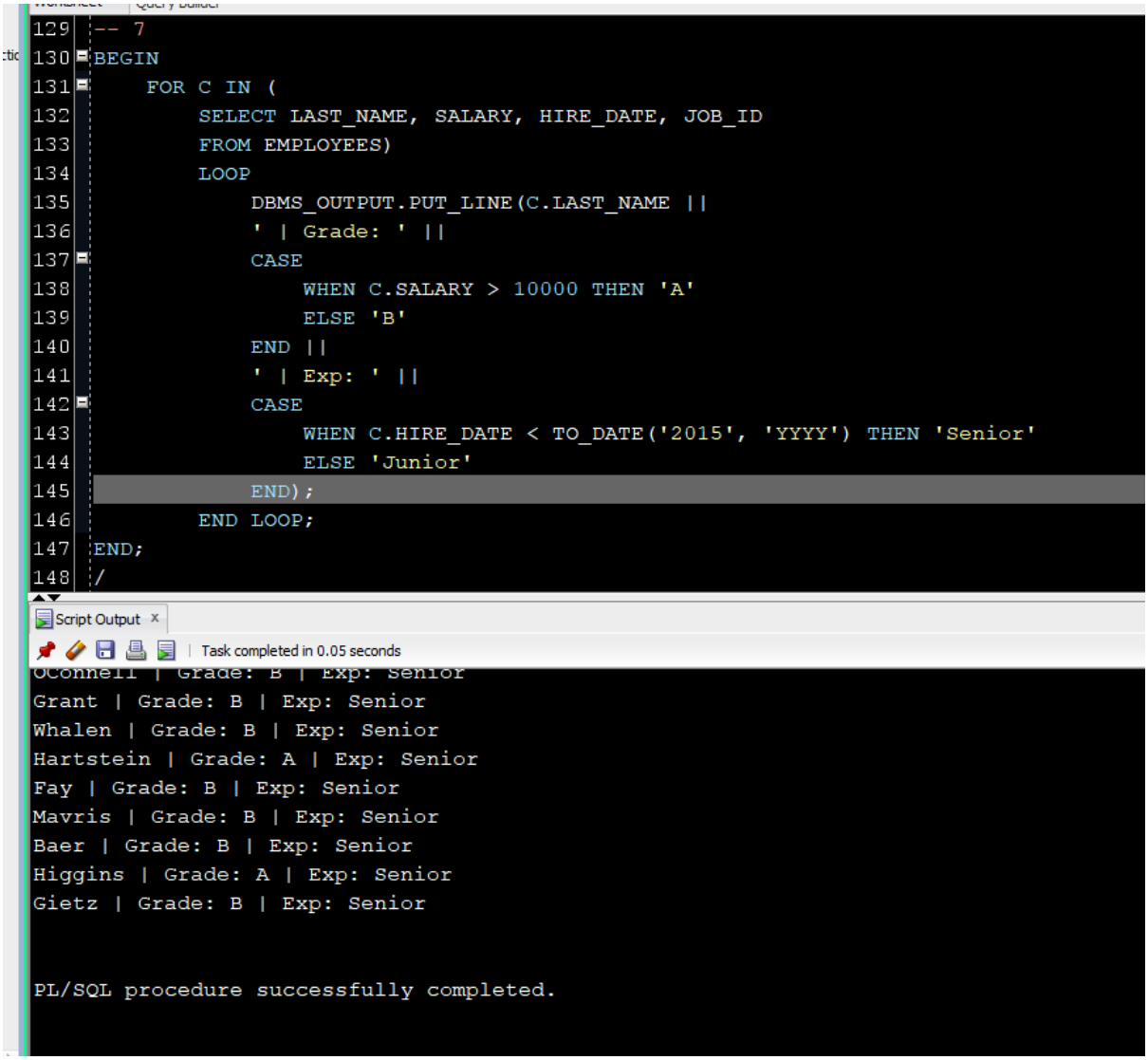
```
-- 6
DECLARE
    TARGET_DEPT EMPLOYEES.DEPARTMENT_ID%TYPE := 100;
BEGIN
    FOR C IN (
        SELECT * FROM EMPLOYEES
        WHERE DEPARTMENT_ID = TARGET_DEPT)
    LOOP
        DBMS_OUTPUT.PUT_LINE('Report: ' || C.LAST_NAME);
        DBMS_OUTPUT.PUT_LINE('Job Class: ' || SUBSTR(C.JOB_ID, 1, 2));
        DBMS_OUTPUT.PUT_LINE('Eligibility: ' ||
            CASE
                WHEN C.SALARY > 7000 THEN 'Qualified'
                ELSE 'Pending'
            END);
        DBMS_OUTPUT.PUT_LINE('Promotion: ' ||
            CASE
                WHEN C.COMMISSION_PCT IS NOT NULL THEN 'Consider'
                ELSE 'N/A'
            END);
        DBMS_OUTPUT.PUT_LINE(' ');
    END LOOP;
END;
```

Script Output

Task completed in 0.041 seconds

```
Report: Popp
Job Class: FI
Eligibility: Pending
Promotion: N/A


PL/SQL procedure successfully completed.
```

7)


-- 7
BEGIN
FOR C IN (
SELECT LAST_NAME, SALARY, HIRE_DATE, JOB_ID
FROM EMPLOYEES)
LOOP
DBMS_OUTPUT.PUT_LINE(C.LAST_NAME ||
' | Grade: ' ||
CASE
WHEN C.SALARY > 10000 THEN 'A'
ELSE 'B'
END ||
' | Exp: ' ||
CASE
WHEN C.HIRE_DATE < TO_DATE('2015', 'YYYY') THEN 'Senior'
ELSE 'Junior'
END);
END LOOP;
END;
/
Script Output
Task completed in 0.05 seconds
OConnell | Grade: B | Exp: Senior
Grant | Grade: B | Exp: Senior
Whalen | Grade: B | Exp: Senior
Hartstein | Grade: A | Exp: Senior
Fay | Grade: B | Exp: Senior
Mavris | Grade: B | Exp: Senior
Baer | Grade: B | Exp: Senior
Higgins | Grade: A | Exp: Senior
Gietz | Grade: B | Exp: Senior
PL/SQL procedure successfully completed.

8)

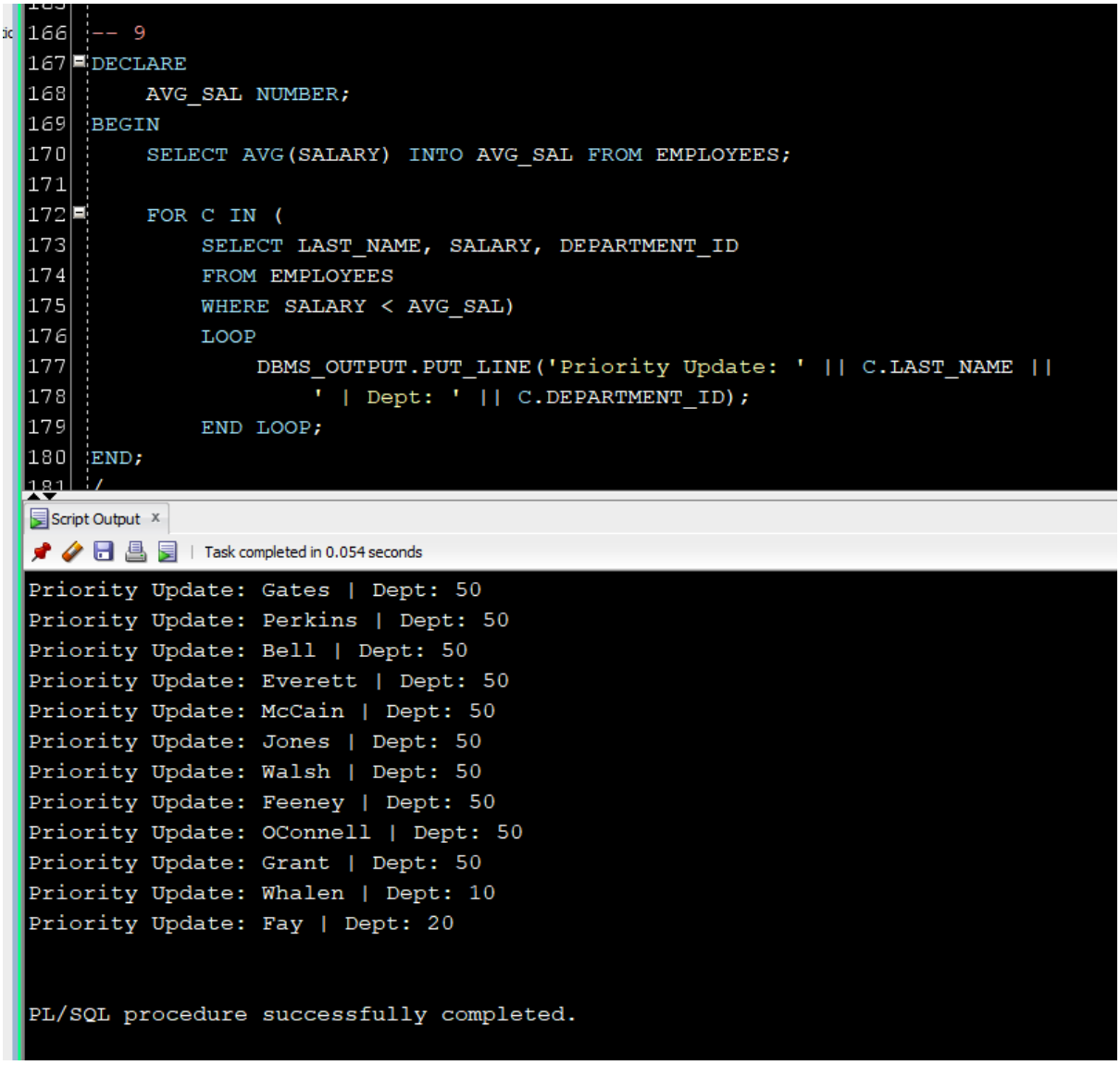
```
149
150 -- 8
151 BEGIN
152     FOR C IN(
153     SELECT LAST_NAME, DEPARTMENT_ID, SALARY, COMMISSION_PCT
154     FROM EMPLOYEES)
155     LOOP
156         IF (C.DEPARTMENT_ID = 80 AND C.COMMISSION_PCT > 0)
157             OR C.SALARY < 5000 THEN
158             DBMS_OUTPUT.PUT_LINE(C.LAST_NAME || ': Eligible for Bonus');
159         ELSE
160             DBMS_OUTPUT.PUT_LINE(C.LAST_NAME || ': No Bonus');
161         END IF;
162     END LOOP;
163 END;
164 /
```

Script Output

Task completed in 0.053 seconds

```
Jones: Eligible for Bonus
Walsh: Eligible for Bonus
Feeney: Eligible for Bonus
OConnell: Eligible for Bonus
Grant: Eligible for Bonus
Whalen: Eligible for Bonus
Hartstein: No Bonus
Fay: No Bonus
Mavris: No Bonus
Baer: No Bonus
Higgins: No Bonus
Gietz: No Bonus


PL/SQL procedure successfully completed.
```

9)



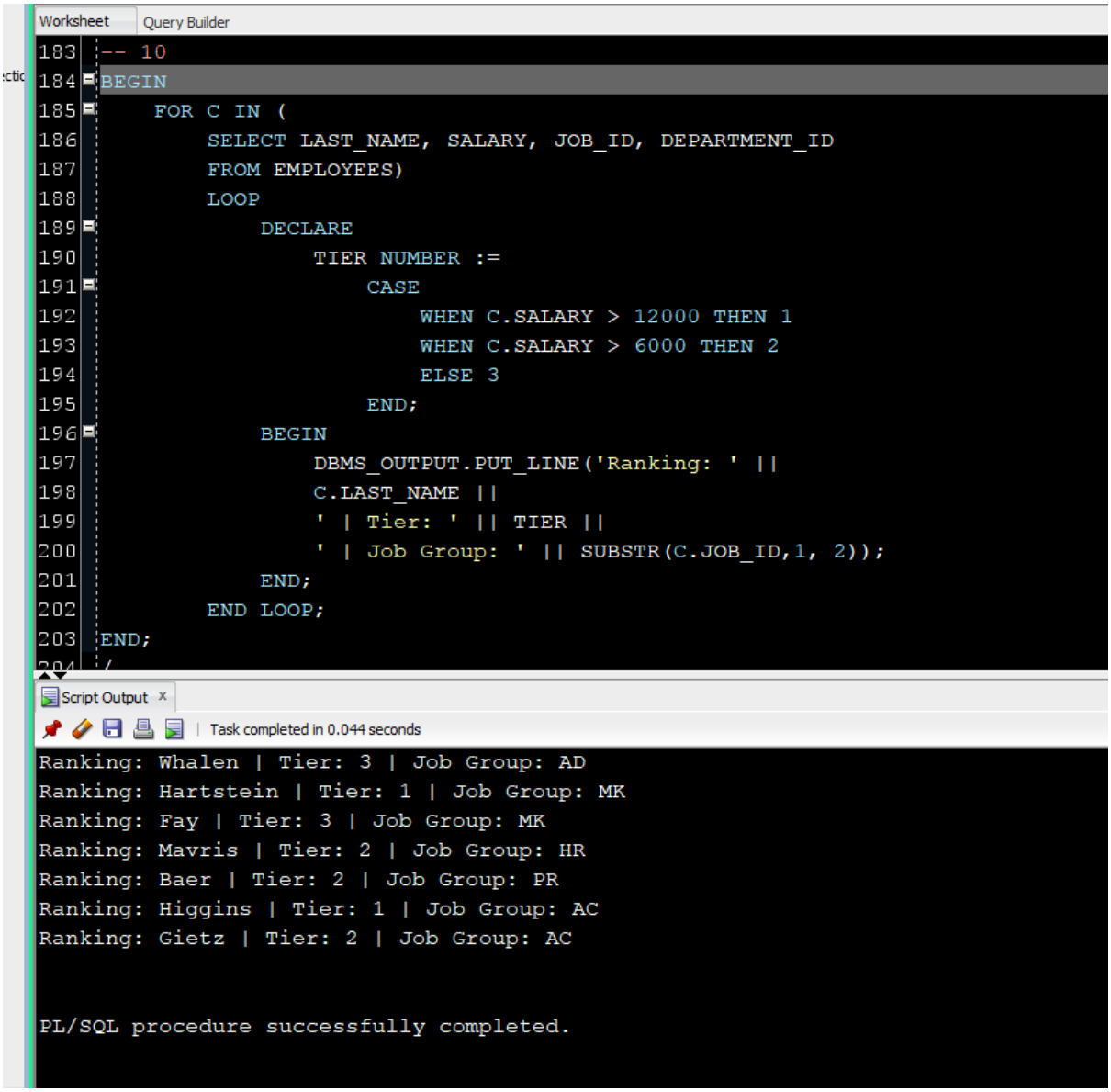
```
166  -- 9
167  DECLARE
168      AVG_SAL NUMBER;
169  BEGIN
170      SELECT AVG(SALARY) INTO AVG_SAL FROM EMPLOYEES;
171
172      FOR C IN (
173          SELECT LAST_NAME, SALARY, DEPARTMENT_ID
174          FROM EMPLOYEES
175          WHERE SALARY < AVG_SAL)
176          LOOP
177              DBMS_OUTPUT.PUT_LINE('Priority Update: ' || C.LAST_NAME ||
178                  ' | Dept: ' || C.DEPARTMENT_ID);
179          END LOOP;
180  END;
181  /
```

Script Output

Task completed in 0.054 seconds

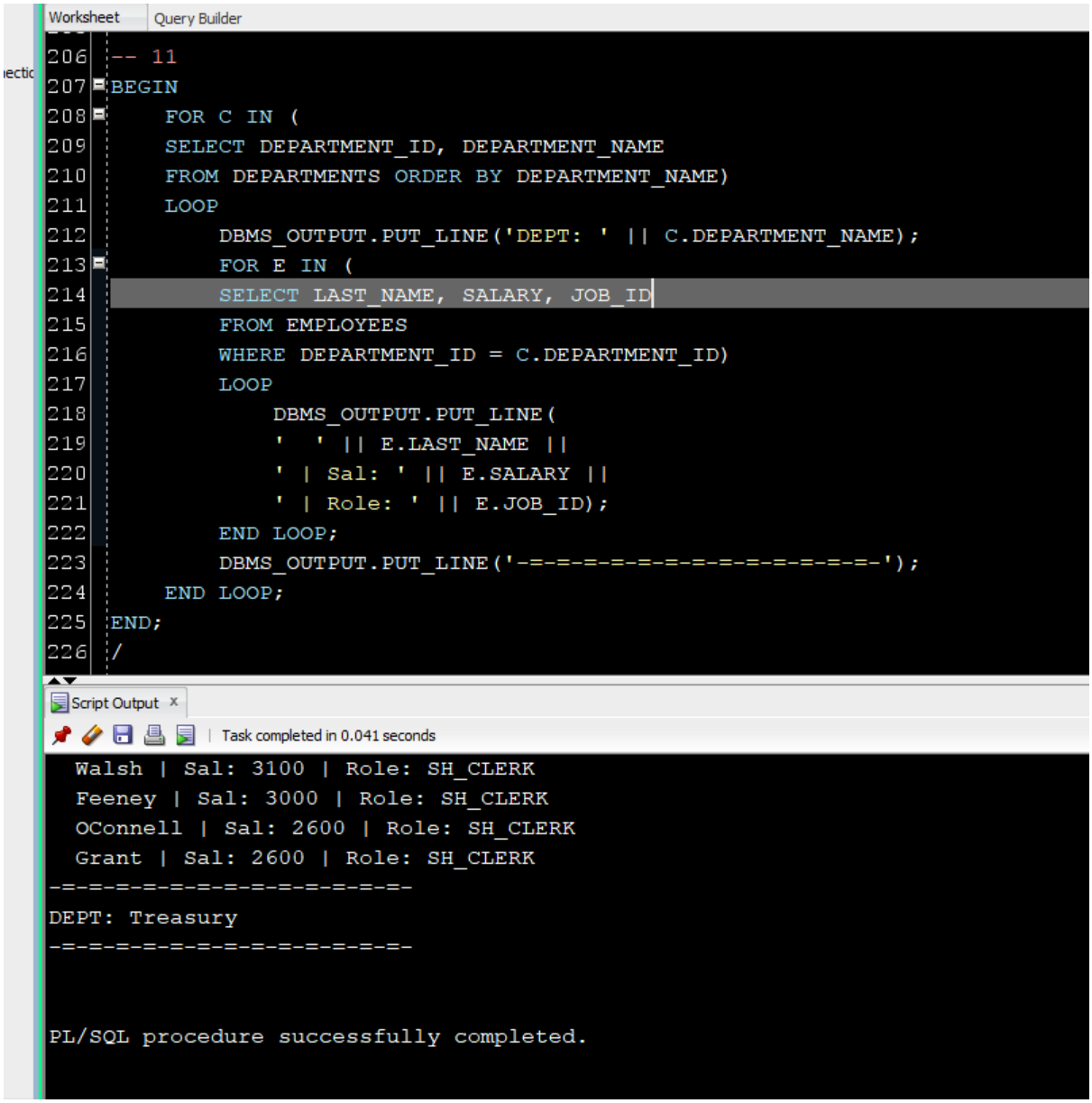
```
Priority Update: Gates | Dept: 50
Priority Update: Perkins | Dept: 50
Priority Update: Bell | Dept: 50
Priority Update: Everett | Dept: 50
Priority Update: McCain | Dept: 50
Priority Update: Jones | Dept: 50
Priority Update: Walsh | Dept: 50
Priority Update: Feeney | Dept: 50
Priority Update: OConnell | Dept: 50
Priority Update: Grant | Dept: 50
Priority Update: Whalen | Dept: 10
Priority Update: Fay | Dept: 20


PL/SQL procedure successfully completed.
```

10)

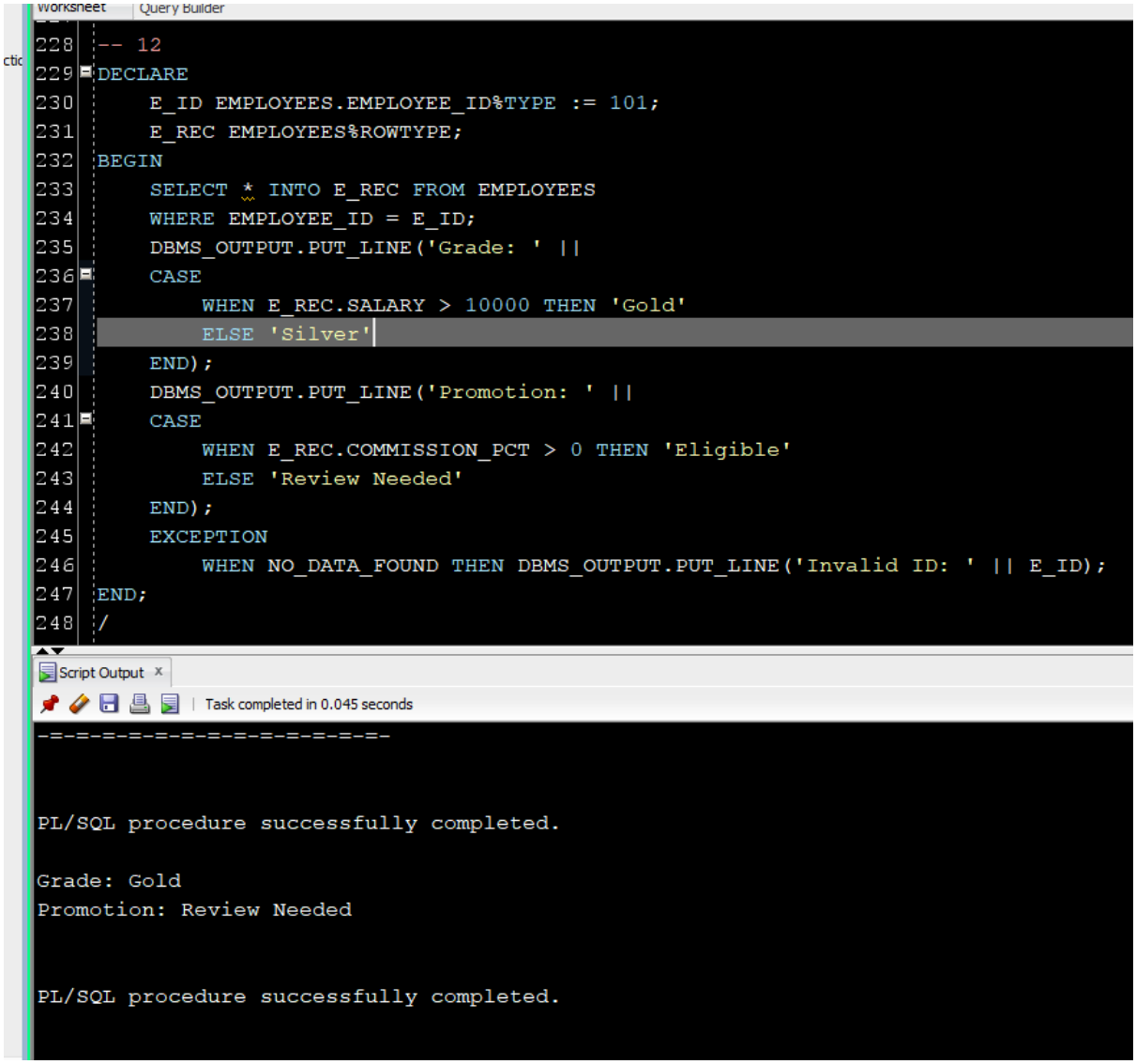


```
Worksheet  Query Builder
183  -- 10
184  BEGIN
185      FOR C IN (
186          SELECT LAST_NAME, SALARY, JOB_ID, DEPARTMENT_ID
187          FROM EMPLOYEES)
188          LOOP
189              DECLARE
190                  TIER NUMBER :=
191                      CASE
192                          WHEN C.SALARY > 12000 THEN 1
193                          WHEN C.SALARY > 6000 THEN 2
194                          ELSE 3
195                      END;
196              BEGIN
197                  DBMS_OUTPUT.PUT_LINE('Ranking: ' ||
198                  C.LAST_NAME ||
199                  ' | Tier: ' || TIER ||
200                  ' | Job Group: ' || SUBSTR(C.JOB_ID,1, 2));
201              END;
202          END LOOP;
203  END;
204  /

Script Output x
Task completed in 0.044 seconds

Ranking: Whalen | Tier: 3 | Job Group: AD
Ranking: Hartstein | Tier: 1 | Job Group: MK
Ranking: Fay | Tier: 3 | Job Group: MK
Ranking: Mavris | Tier: 2 | Job Group: HR
Ranking: Baer | Tier: 2 | Job Group: PR
Ranking: Higgins | Tier: 1 | Job Group: AC
Ranking: Gietz | Tier: 2 | Job Group: AC


PL/SQL procedure successfully completed.
```

11)


Worksheet
Query Builder
206 -- 11
207 BEGIN
208 FOR C IN (
209 SELECT DEPARTMENT_ID, DEPARTMENT_NAME
210 FROM DEPARTMENTS ORDER BY DEPARTMENT_NAME)
211 LOOP
212 DBMS_OUTPUT.PUT_LINE('DEPT: ' || C.DEPARTMENT_NAME);
213 FOR E IN (
214 SELECT LAST_NAME, SALARY, JOB_ID
215 FROM EMPLOYEES
216 WHERE DEPARTMENT_ID = C.DEPARTMENT_ID)
217 LOOP
218 DBMS_OUTPUT.PUT_LINE(
219 ' ' || E.LAST_NAME ||
220 ' | Sal: ' || E.SALARY ||
221 ' | Role: ' || E.JOB_ID);
222 END LOOP;
223 DBMS_OUTPUT.PUT_LINE('-=-=-=-=-=-=-=-=-=-=-=-=-=-');
224 END LOOP;
225 END;
226 /
Script Output
Task completed in 0.041 seconds
Walsh | Sal: 3100 | Role: SH_CLERK
Feeney | Sal: 3000 | Role: SH_CLERK
OConnell | Sal: 2600 | Role: SH_CLERK
Grant | Sal: 2600 | Role: SH_CLERK
-=-=-=-=-=-=-=-=-=-=-=-=-=-
DEPT: Treasury
-=-=-=-=-=-=-=-=-=-=-=-=-=-
PL/SQL procedure successfully completed.

12)



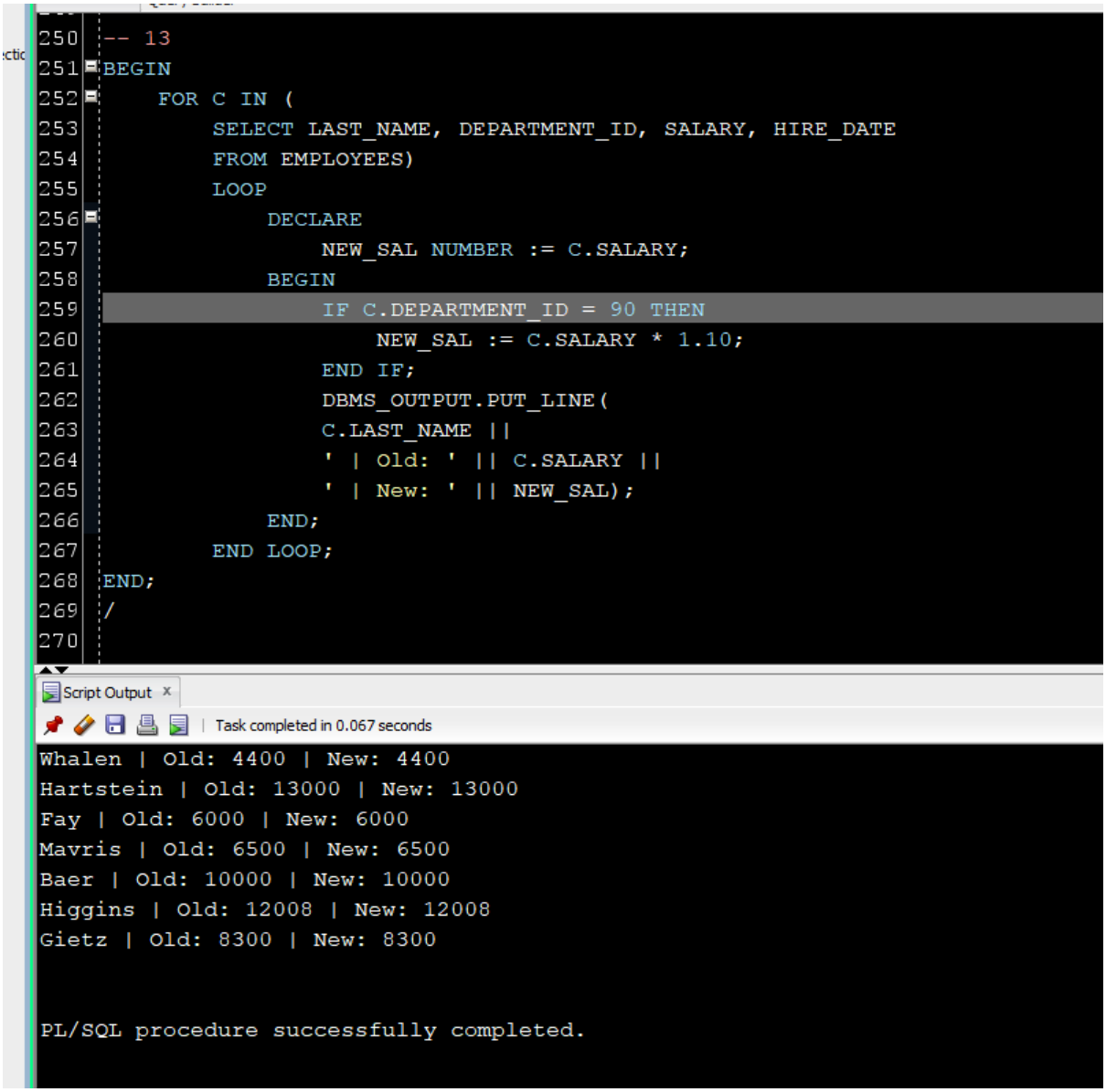
```
-- 12
DECLARE
    E_ID EMPLOYEES.EMPLOYEE_ID%TYPE := 101;
    E_REC EMPLOYEES%ROWTYPE;
BEGIN
    SELECT * INTO E_REC FROM EMPLOYEES
    WHERE EMPLOYEE_ID = E_ID;
    DBMS_OUTPUT.PUT_LINE('Grade: ' ||
    CASE
        WHEN E_REC.SALARY > 10000 THEN 'Gold'
        ELSE 'Silver'
    END);
    DBMS_OUTPUT.PUT_LINE('Promotion: ' ||
    CASE
        WHEN E_REC.COMMISSION_PCT > 0 THEN 'Eligible'
        ELSE 'Review Needed'
    END);
    EXCEPTION
        WHEN NO_DATA_FOUND THEN DBMS_OUTPUT.PUT_LINE('Invalid ID: ' || E_ID);
END;
/
```

Script Output

Task completed in 0.045 seconds

```
-=-=-=-=-=-=-=-=-=-=-=-=-

PL/SQL procedure successfully completed.

Grade: Gold
Promotion: Review Needed


PL/SQL procedure successfully completed.
```

13)



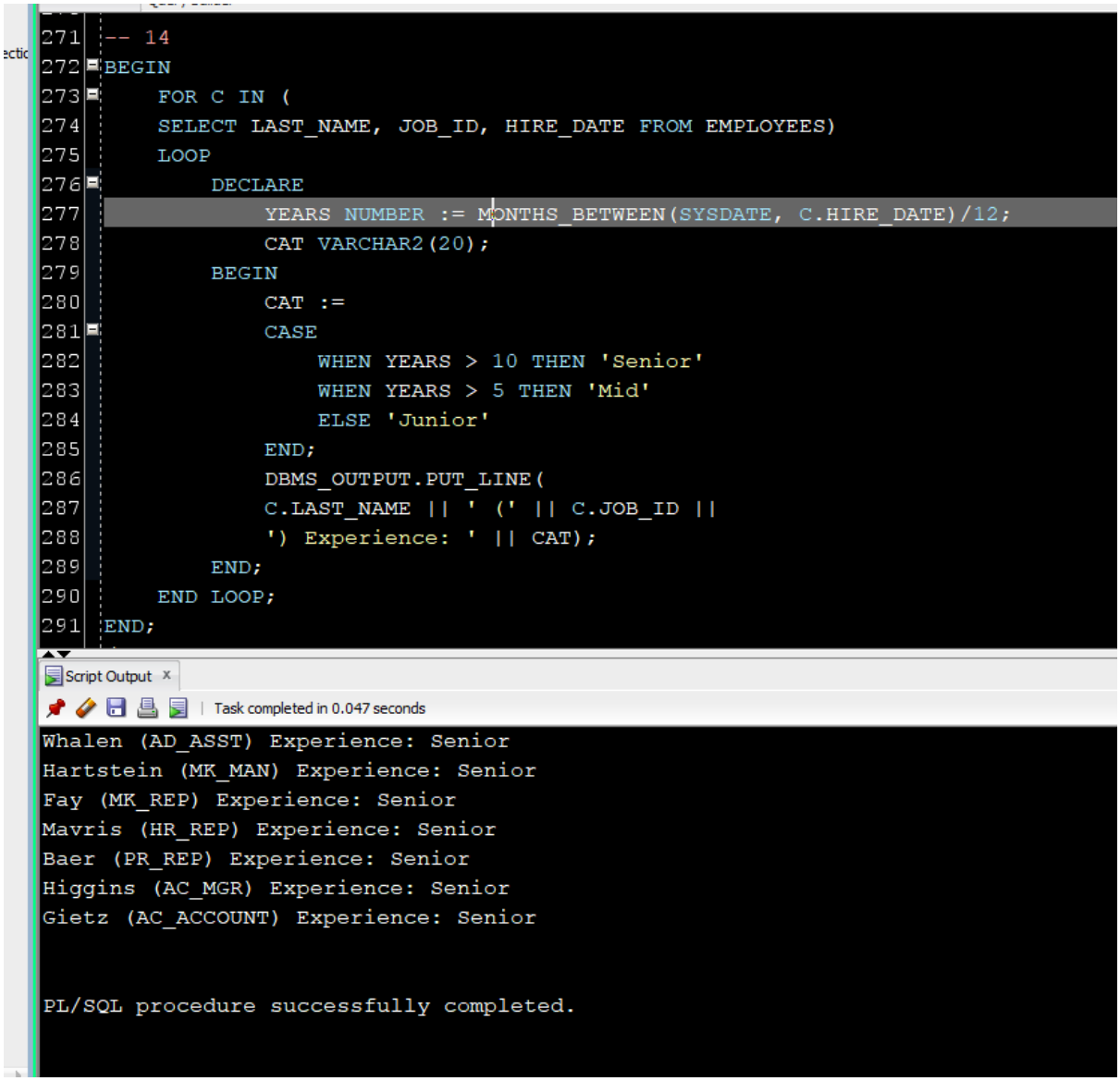
```
-- 13
BEGIN
    FOR C IN (
        SELECT LAST_NAME, DEPARTMENT_ID, SALARY, HIRE_DATE
        FROM EMPLOYEES)
        LOOP
            DECLARE
                NEW_SAL NUMBER := C.SALARY;
            BEGIN
                IF C.DEPARTMENT_ID = 90 THEN
                    NEW_SAL := C.SALARY * 1.10;
                END IF;
                DBMS_OUTPUT.PUT_LINE(
                C.LAST_NAME ||
                ' | Old: ' || C.SALARY ||
                ' | New: ' || NEW_SAL);
            END;
        END LOOP;
END;
/
```

Script Output

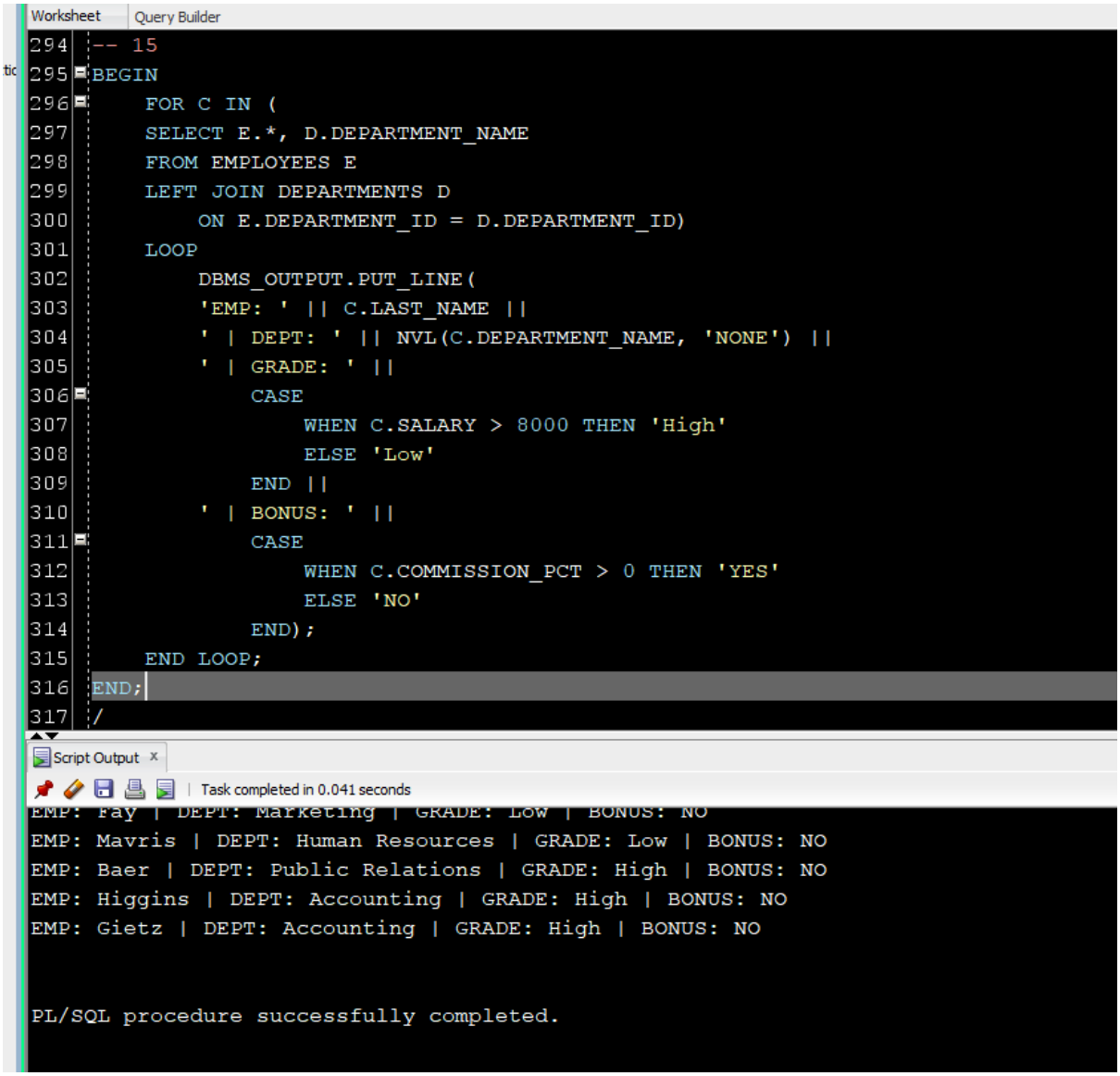
Task completed in 0.067 seconds

```
Whalen | Old: 4400 | New: 4400
Hartstein | Old: 13000 | New: 13000
Fay | Old: 6000 | New: 6000
Mavris | Old: 6500 | New: 6500
Baer | Old: 10000 | New: 10000
Higgins | Old: 12008 | New: 12008
Gietz | Old: 8300 | New: 8300


PL/SQL procedure successfully completed.
```

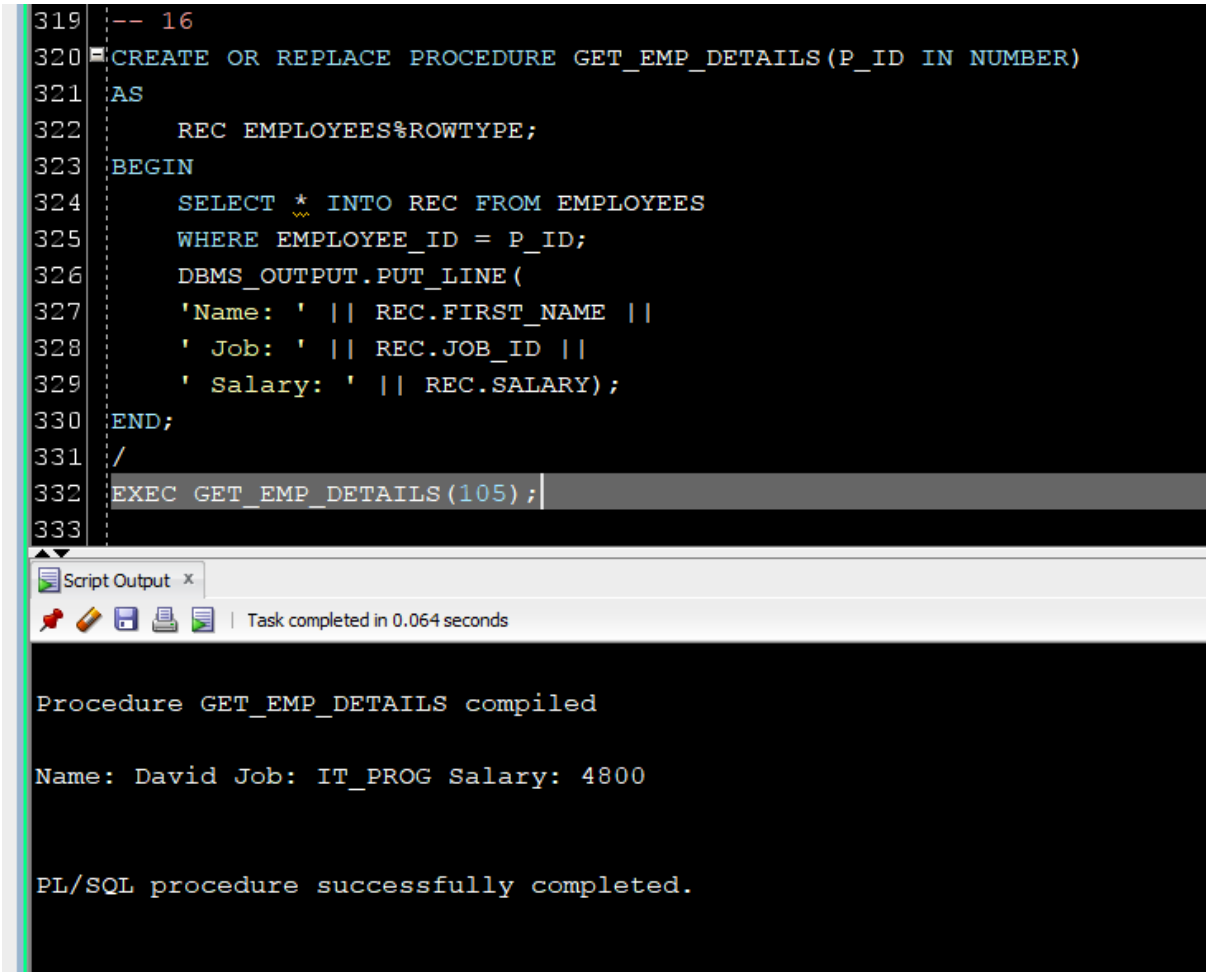
14)


-- 14
BEGIN
FOR C IN (
SELECT LAST_NAME, JOB_ID, HIRE_DATE FROM EMPLOYEES)
LOOP
DECLARE
YEARS NUMBER := MONTHS_BETWEEN(SYSDATE, C.HIRE_DATE)/12;
CAT VARCHAR2(20);
BEGIN
CAT :=
CASE
WHEN YEARS > 10 THEN 'Senior'
WHEN YEARS > 5 THEN 'Mid'
ELSE 'Junior'
END;
DBMS_OUTPUT.PUT_LINE(
C.LAST_NAME || ' (' || C.JOB_ID ||
') Experience: ' || CAT);
END;
END LOOP;
END;
Script Output
Task completed in 0.047 seconds
Whalen (AD_ASST) Experience: Senior
Hartstein (MK_MAN) Experience: Senior
Fay (MK_REP) Experience: Senior
Mavris (HR_REP) Experience: Senior
Baer (PR_REP) Experience: Senior
Higgins (AC_MGR) Experience: Senior
Gietz (AC_ACCOUNT) Experience: Senior
PL/SQL procedure successfully completed.

15)


Worksheet
Query Builder
-- 15
BEGIN
FOR C IN (
SELECT E.*, D.DEPARTMENT_NAME
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.DEPARTMENT_ID = D.DEPARTMENT_ID)
LOOP
DBMS_OUTPUT.PUT_LINE(
'EMP: ' || C.LAST_NAME ||
' | DEPT: ' || NVL(C.DEPARTMENT_NAME, 'NONE') ||
' | GRADE: ' ||
CASE
WHEN C.SALARY > 8000 THEN 'High'
ELSE 'Low'
END ||
' | BONUS: ' ||
CASE
WHEN C.COMMISSION_PCT > 0 THEN 'YES'
ELSE 'NO'
END);
END LOOP;
END;
/
Script Output
Task completed in 0.041 seconds
EMP: Fay | DEPT: Marketing | GRADE: Low | BONUS: NO
EMP: Mavris | DEPT: Human Resources | GRADE: Low | BONUS: NO
EMP: Baer | DEPT: Public Relations | GRADE: High | BONUS: NO
EMP: Higgins | DEPT: Accounting | GRADE: High | BONUS: NO
EMP: Gietz | DEPT: Accounting | GRADE: High | BONUS: NO
PL/SQL procedure successfully completed.

16)


319 -- 16
320 CREATE OR REPLACE PROCEDURE GET_EMP_DETAILS(P_ID IN NUMBER)
321 AS
322 REC EMPLOYEES%ROWTYPE;
323 BEGIN
324 SELECT * INTO REC FROM EMPLOYEES
325 WHERE EMPLOYEE_ID = P_ID;
326 DBMS_OUTPUT.PUT_LINE(
327 'Name: ' || REC.FIRST_NAME ||
328 ' Job: ' || REC.JOB_ID ||
329 ' Salary: ' || REC.SALARY);
330 END;
331 /
332 EXEC GET_EMP_DETAILS(105);
333
Script Output
Task completed in 0.064 seconds
Procedure GET_EMP_DETAILS compiled
Name: David Job: IT_PROG Salary: 4800
PL/SQL procedure successfully completed.

17)

```
334  -- 17
335  CREATE OR REPLACE PROCEDURE RAISE_SALARY(
336      P_ID IN NUMBER, P_PERCENT IN NUMBER) AS
337      E_NAME EMPLOYEES.LAST_NAME%TYPE;
338      NEW_SAL NUMBER;
339  BEGIN
340      UPDATE EMPLOYEES SET SALARY = SALARY * (1 * P_PERCENT / 100)
341      WHERE EMPLOYEE_ID = P_ID
342          RETURNING LAST_NAME, SALARY INTO E_NAME, NEW_SAL;
343      DBMS_OUTPUT.PUT_LINE(
344      E_NAME || ' Updated Salary: ' || NEW_SAL);
345  END;
346  /
347  EXEC RAISE_SALARY(150, 12);
348
349  -- 18
```

Script Output

Task completed in 0.056 seconds

```
Name: David Job: IT_PROG Salary: 4800


PL/SQL procedure successfully completed.

Tucker Updated Salary: 1200


PL/SQL procedure successfully completed.
```

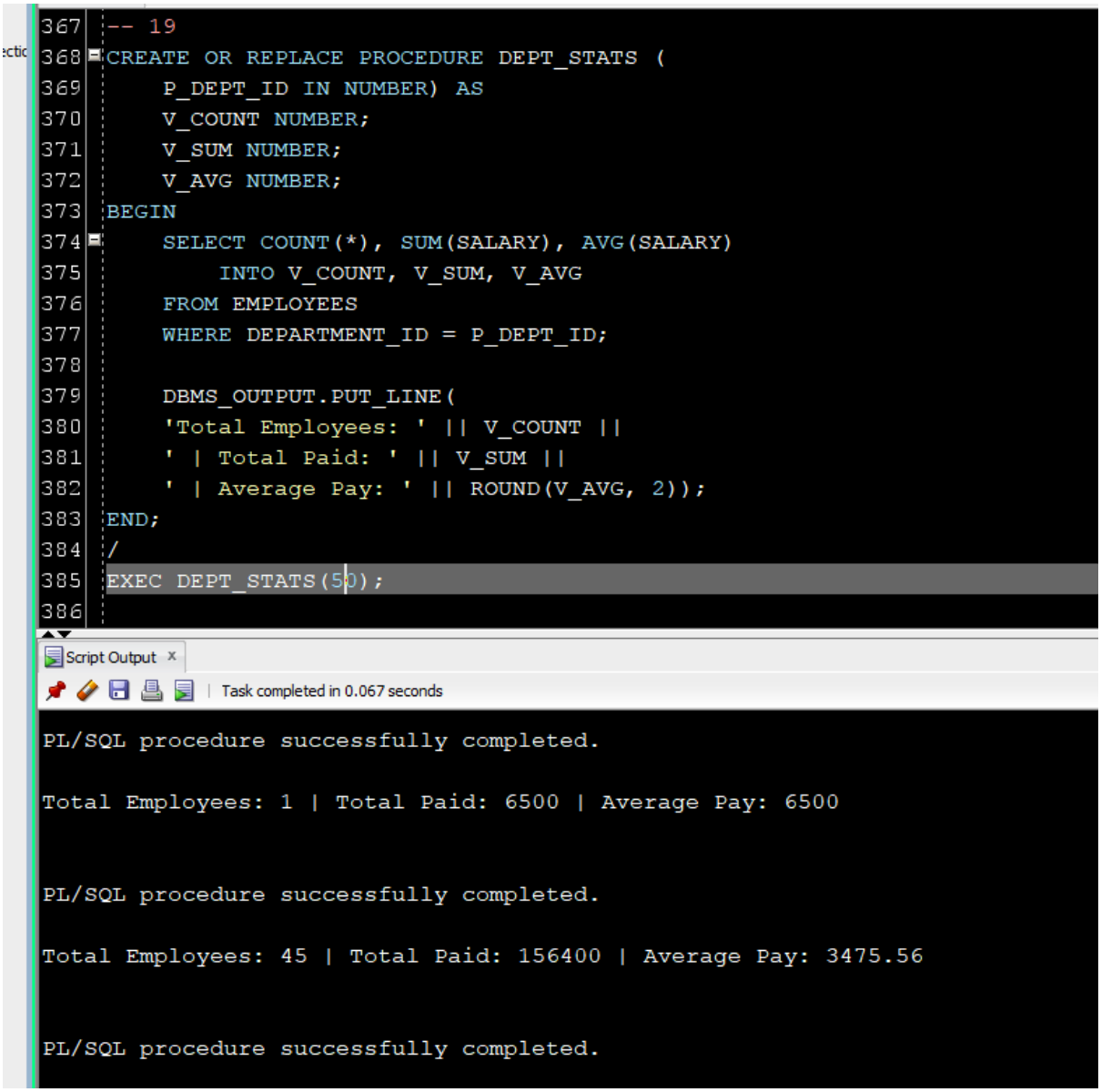
```
349 -- 18
350 CREATE OR REPLACE PROCEDURE LIST_DEPT_EMPS(
351     P_DEPT_ID IN NUMBER) AS
352 BEGIN
353     FOR C IN (
354     SELECT LAST_NAME, JOB_ID, SALARY
355     FROM EMPLOYEES
356     WHERE DEPARTMENT_ID = P_DEPT_ID)
357     LOOP
358         DBMS_OUTPUT.PUT_LINE(
359         C.LAST_NAME ||
360         ' | ' || C.JOB_ID ||
361         ' | ' || C.SALARY);
362     END LOOP;
363 END;
364 /
365 EXEC LIST_DEPT_EMPS(50);
```

Script Output

Task completed in 0.064 seconds

```
Bell | SH_CLERK | 4000
Everett | SH_CLERK | 3900
McCain | SH_CLERK | 3200
Jones | SH_CLERK | 2800
Walsh | SH_CLERK | 3100
Feeney | SH_CLERK | 3000
OConnell | SH_CLERK | 2600
Grant | SH_CLERK | 2600


PL/SQL procedure successfully completed.
```

19)



```
-- 19
CREATE OR REPLACE PROCEDURE DEPT_STATS (
    P_DEPT_ID IN NUMBER) AS
    V_COUNT NUMBER;
    V_SUM NUMBER;
    V_AVG NUMBER;
BEGIN
    SELECT COUNT(*), SUM(SALARY), AVG(SALARY)
        INTO V_COUNT, V_SUM, V_AVG
    FROM EMPLOYEES
    WHERE DEPARTMENT_ID = P_DEPT_ID;

    DBMS_OUTPUT.PUT_LINE(
    'Total Employees: ' || V_COUNT ||
    ' | Total Paid: ' || V_SUM ||
    ' | Average Pay: ' || ROUND(V_AVG, 2));
END;
/
EXEC DEPT_STATS(50);
```

Script Output

Task completed in 0.067 seconds

```
PL/SQL procedure successfully completed.

Total Employees: 1 | Total Paid: 6500 | Average Pay: 6500


PL/SQL procedure successfully completed.

Total Employees: 45 | Total Paid: 156400 | Average Pay: 3475.56


PL/SQL procedure successfully completed.
```

20)

```
387  -- 20
388  CREATE OR REPLACE PROCEDURE LIST_BY_JOB(
389      P_JOB_ID IN VARCHAR2) AS
390  BEGIN
391      FOR C IN (
392      SELECT E.LAST_NAME, D.DEPARTMENT_NAME, E.SALARY
393      FROM EMPLOYEES E
394      JOIN DEPARTMENTS D
395          ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
396      WHERE E.JOB_ID = P_JOB_ID)
397      LOOP
398          DBMS_OUTPUT.PUT_LINE(
399          C.LAST_NAME || ' in ' ||
400          C.DEPARTMENT_NAME || ': $' ||
401          C.SALARY);
402      END LOOP;
403  END;
404  /
405  EXEC LIST_BY_JOB('SA_REP');
```

Script Output

Task completed in 0.051 seconds

```
Fox in Sales: $9600
Smith in Sales: $7400
Bates in Sales: $7300
Kumar in Sales: $6100
Abel in Sales: $11000
Hutton in Sales: $8800
Taylor in Sales: $8600
Livingston in Sales: $8400
Johnson in Sales: $6200


PL/SQL procedure successfully completed.
```